CODEALPHA • INTERNET OF THINGS

AUTOMATIC LIGHT CONTROL USING LDR

Task 2 — Sensor-Based Simulation

Student: Niharika Varshitha T.

Platform: Tinkercad Circuits • Board: Arduino Uno R3

SIMULATED PROJECT PRESENTATION

1. Objective

To design and simulate an automatic light control system using an LDR sensor and Arduino Uno. The system detects light intensity and controls an LED according to the sensor reading.

SENSE

LDR detects light

PROCESS

Arduino checks value

RESPOND

LED turns ON or OFF

2. Components Used

Arduino Uno R3

Reads the sensor and controls the LED

LDR / Photoresistor

Senses surrounding light

LED

Shows the system output

1 kΩ resistor

Limits current through the LED

10 kΩ resistor

Forms a voltage divider with LDR

Breadboard & jumper wires

Connect the components

Tinkercad Circuits

Online simulation environment

3. Circuit Connections

INSERT Task2_Circuit.png

Replace this box with your saved screenshot

- LDR output junction → A0
- LED control pin → D9
- 1 kΩ resistor in series with LED
- 10 kΩ resistor from LDR junction to GND
- LDR connected to 5V
- Common ground (GND)

4. Working Principle

1	**LDR senses light**	Sensor produces an analog reading.
2	**Arduino reads A0**	The value is checked repeatedly.
3	**Compare with 500**	The program uses 500 as its threshold.
4	**Control the LED**	Below 500: ON • 500 or above: OFF.

5. Arduino Program

```
const int ldrPin = A0;
const int ledPin = 9;
const int threshold = 500;

void setup() {
  pinMode(ledPin, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int lightLevel = analogRead(ldrPin);

  if (lightLevel < threshold) {
    digitalWrite(ledPin, HIGH);
  } else {
    digitalWrite(ledPin, LOW);
  }

  Serial.println(lightLevel);
  delay(200);
}
```

INSERT Task2_Code.png

Replace this box with your saved screenshot

Threshold = 500

LED pin = 9 • LDR pin = A0

6. Code Explanation

A0	Analog input used to read the LDR value.
Pin 9	Digital output used to control the LED.
Threshold 500	Reference value for switching the LED.
analogRead()	Reads the current sensor value.
digitalWrite()	Sets the LED output HIGH or LOW.
Serial.println()	Displays readings in the Serial Monitor.
delay(200)	Waits 200 milliseconds between readings.

7. Simulation Result — LED ON

INSERT Task2_LED_ON.png

Replace this box with your saved screenshot

LOW-LIGHT TEST

Example reading

54 (below 500)

LED STATUS: ON

The LED glows when the reading is below 500.

8. Simulation Result — LED OFF

INSERT Task2_LED_OFF.png

Replace this box with your saved screenshot

HIGH-READING TEST

Example reading

973 (above 500)

LED STATUS: OFF

The LED switches off when the reading is above 500.

9. Overall Result

The circuit responded to sensor readings and switched the LED according to the programmed threshold.

LDR reading < 500	LDR reading = 500	LDR reading > 500
Low-light test	At threshold	High-reading test
LED ON	LED OFF	LED OFF

Note: A reading equal to 500 follows the ELSE branch, so the LED is OFF.

CONCLUSION

The project demonstrates a simple sensor-based automation system. The LDR provides an input reading, and the Arduino processes it to control an LED automatically.

THANK YOU

CodeAlpha IoT Internship • Task 2